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Precambrian Research 129 (2004) 251–270
Precambrian Research
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Early Proterozoic geomagnetic field in western Laurentia: implications for paleolatitudes, local rotations and stratigraphy☆

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Abstract

The east-west trending, post-orogenic subvertical Cleaver Dykes from the Great Bear Magmatic Arc are dated by the U–Pb (baddeleyite) method to be 1740 +5/−4 Ma. They have a mean direction of magnetization of *D*, $I = 136.5$ and 57.4°, $k = 64$, $\alpha_{95} = 4.5°$, based on samples from 17 diabase dykes, a mean location 65.7°N, 118.03°W, and a corresponding paleopole at 19.4°N, 83.3°W, $A_{95} = 6.1°$. Studies of baked contacts, country rocks distant from dykes, and later intrusions that cut them, show that the dykes became magnetized soon after intrusion. The paleopole is not significantly different from that obtained from the 1759 ± 1 Ma post-orogenic Jan Lake Granite from the Hudsonian Orogen in southeast Saskatchewan. These paleopoles fall among a cluster previously determined from post-orogenic overprints in a wide variety of rocks and geological settings across Laurentia. Their mean is 20.8°N, 94.5°W, $A_{95} = 5.2°$. Collectively they record a widespread remagnetization episode following the Hudsonian Orogeny, during which the polarity of the field remained constant. This single polarity interval (the Cleaver Superchron) has a mean age of approximately 1747 Ma, and its duration is at least 13 million years but is probably much longer. With further refinement, it could become a global statigraphical marker.

There are 18 paleopoles from red beds and igneous rocks which record the paleofield in western Laurentia mainly over an extended interval we estimate to be at least 100 million years between about 1960 and 1830 Ma during the Hudsonian Orogeny. Over this interval, western Laurentia remained at a low and essentially constant latitude. Reversals of the geomagnetic field occurred opposed magnetizations being of approximately equal frequency; there are numerous magnetochrons which could, in future, assist in regional stratigraphic correlations. On the basis of six studies, we estimate that the reference paleopole for this interval is at 1.1°S, 71.7°W, $A_{95} = 12.1°$ ($N = 6$). Paleopoles from the other 12 studies are spread over a 100° arc indicating large variable rotations about local vertical axes which probably resulted from transcurrent faulting caused by indentation of the bordering Archean Slave and Superior cratons during the Hudsonian Orogeny.

These reference paleopoles show that during the Hudsonian Orogeny, western Laurentia was in paleolatitudes of 15–30°. During post-Hudsonian time (ca. 1750 Ma) paleolatitudes were about 20° higher. Paleowind directions inferred by others, indicate that Laurentia during these intervals was in the northern hemisphere.
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Keywords: Paleomagnetism; Cleaver Dykes; Baddeleyite age; Rotations; Magnetostratigraphy

1. Introduction

In 1967, Alan Donaldson and the late John McGlynn proposed to the first author that they make paleomagnetic surveys of basinal sequences in the western Canadian Shield. It is therefore appropriate

☆ Geological Survey of Canada publication number 2003013.

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0301-9268/$ – see front matter © 2003 Published by Elsevier B.V.
doi:10.1016/j.precamres.2003.10.002

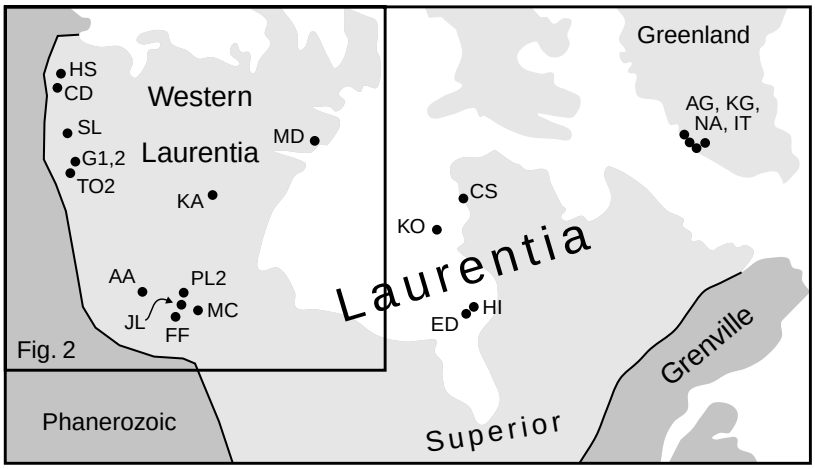


Fig. 1. Sketch map of Laurentia. Greenland is reconstructed to its Precambrian position by closing the Labrador Sea. Sampling localities used to estimate the post-Hudsonian geomagnetic field labeled as in Fig. 11.

that we should summarize, in this publication in Al's honour, the present status of such studies and assess their significance.

Laurentia is an aggregate of small continents that were assembled during the Proterozoic Era. We are concerned primarily with western Laurentia, the presently exposed part of Laurentia north of the Trans-Hudson suture and west of Hudson Bay (Figs. 1–2). Western Laurentia comprises the Archean Slave, Rae and Hearne Provinces which were assembled during the Paleoproterozoic beginning at approximately 1950 Ma according to Hoffman (1989). The Rae Province extends east of Hudson Bay and south as far as the Grenville Front, but only reconnaissance paleomagnetic data are available from this prolongation (Park, 1977), which is not therefore considered. The Trans-Hudson Orogen (Fig. 2) was formed between about 1900 and 1800 Ma in a contracting ocean whose final collapse culminated in the collision and suturing of the Superior and the Slave, Rae and Hearne (formerly the Churchill Province) cratons (Gibb and Walcott, 1971; Thomas and Gibb, 1983; Hoffman, 1989). The Wathaman-Chipewayan Arc to the southeast and the Great Bear Arc to the northwest were added at approximately 1850 Ma. The entire episode (ca. 1950–1800), during which the Slave, Rae and Hearne and Superior cratons were assembled, we refer to as the Hudsonian Orogeny, as was originally envisaged by Stockwell (1973). Rocks formed during this episode we describe as syn-Hudsonian. Principally, they are the Goulburn, Coronation, Dubawnt, and Great Slave Supergroups which occur within or marginal to the Slave and Rae and Hearne cratons. Following the Hudsonian Orogeny, little-disturbed sedimentary rocks of the Athabasca, Thelon, and Amundsen Basins were deposited and we describe them as post-Hudsonian (Fig. 2).

Paleomagnetic and geochronological studies together provide records of paleolatitude, of the direction of the paleomeridian (the paleodirection of the paleopole from which tectonic rotations are calculated), and of the polarity of the ancient geomagnetic field as a function of time. In this paper, we are concerned with such records for western Laurentia during the latter part of the Early Proterozoic. The availability of data ultimately depends on the exposure of suitable rocks, and the effect has been that the data group themselves into two broad intervals of time corresponding to the syn- and post-Hudsonian episodes just mentioned. For these intervals, we describe the polarity characteristics of the paleofield, construct paleolatitude maps of western Laurentia, and comment on their tectonic, stratigraphic and climatic significance.

The geomagnetic field in the Phanerozoic is characterized by reversals of polarity, the field being either normal (the main geocentric dipole field being directed along the axis of rotation towards the south

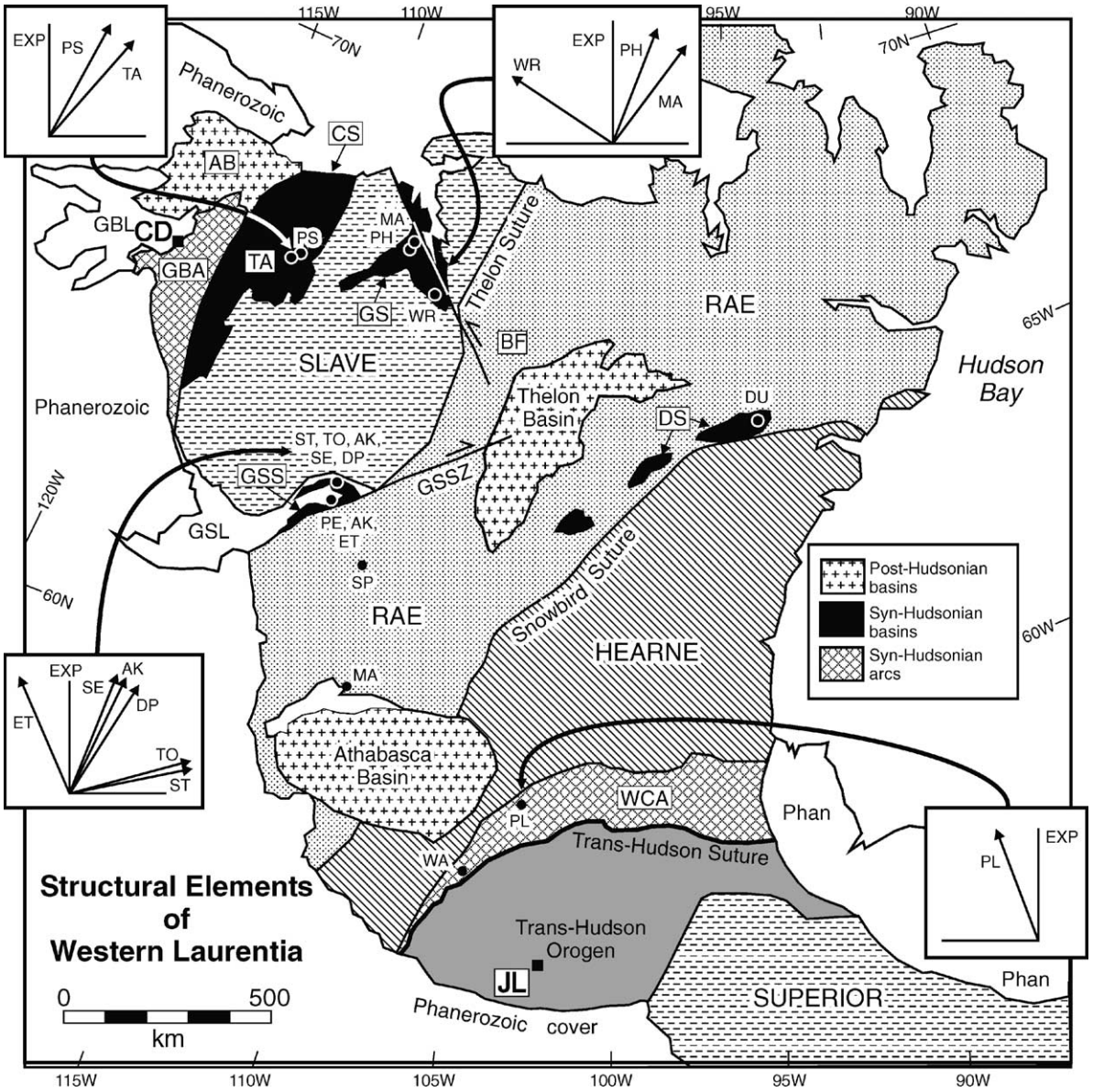


Fig. 2. Sketch map of western Laurentia showing sampling localities of syn-Hudsonian rocks labeled as in Table 3 and Fig. 12. Insets show rotations relative to the reference field given in Section 6. The expected direction (EXP) of the horizontal component of the mean paleofield direction is oriented arbitrarily to the top of the map. Rotations (RRs of Table 3) are the angles between labeled arrows and EXP. AB: Amundsen Basin; BF: Bathurst Inlet; CS: Coronation; DS: Dubawnt; GS: Goulburn; and GSS: Great Slave Supergroups; GBA: Great Bear Arc; GBL: Great Bear Lake; GSL: Great Slave Lake; GSSZ: Great Slave Shear Zone; WCA: Wathaman-Chipewyan Arc.

geographic pole) or reversed (the opposite sense). Because of gaps in the record, the Precambrian paleopoles described herein cannot be traced unambiguously to the Phanerozoic paleopole paths although attempts have been made to do so. It is not possible therefore to designate Proterozoic magnetizations as definitely normal or reversed in the sense that they are used in the Phanerozoic. However, in order to make clear that the phenomenon is the same, we retain the usual terminology but in inverted commas,

and we adopt the convention that magnetizations that are directed generally to the south with downward inclination are described as “normal” and magnetizations directed generally to the north with upward inclination as “reversed”. We shall argue that these are in fact the true polarity designation. We begin by describing new data from the Cleaver Dykes.

2. Relationships and U–Pb geochronology of Cleaver Dykes

Field relationships are shown schematically in Fig. 3. Resting unconformably on the Hottah Terrane (an arc-related sequence extensively deformed and metamorphosed about 1900 Ma, Hoffman, 1989) is the Labine Group (~1870 Ma) with associated intrusions of the Great Bear Magmatic Arc (GBA, Fig. 3; Hildebrand et al., 1987). It comprises several kilometers of lacustrine and fluvial sedimentary rocks, tuffs, lavas, breccias, and massive plutons, and is gently tilted and folded but little metamorphosed (sub-greenschist facies). The Labine Group and associated plutons are intruded by a diabase dyke swarm, the Cleaver Dykes, the oldest post-orogenic intrusions (Cleaver diabase of Hoffman, 1982; Hildebrand, 1983a,b). Unconformably, overlying the Labine Group and truncating the Cleaver Dykes is the sedimentary Hornby Bay Group. The Western Channel Diabase (sills and dykes ~1400 Ma) intrudes the Labine Group and the lower part of the Hornby Bay Group. The Narakay Volcanic Complex (1663 ± 8 Ma) occurs in the upper part of the Hornby Bay Group (Bowring and Ross, 1985). Huge sills representative of the late Proterozoic Franklin Intrusions (locally called the Hottah Gabbro 780 Ma) are the last intrusions.

The Cleaver Dykes are sub-vertical, are typically 5–20 m wide, and have sharp contacts. They trend between east-west and east-southeast to west-northwest. They are typically sub-ophitic to ophitic, with euhedral labradorite phenocrysts (Hildebrand, 1984).

About 15 kg of coarse diabase was collected by Dr. Sunil Gandhi from a thick Cleaver Dyke intruded into sedimentary and volcanic rocks of the Terra Formation on the western flank of the Great Bear Arc (118°07′W, 65°36.2′N). Crushing, grinding, Wilfley table and heavy liquid techniques followed the methods outlined in Hamilton et al. (2001), and resulted in a modest concentration of baddeleyite crystals. Most were short, thin, honey-brown striated blades and blade fragments, typically 10 μm × 40 μm × 80 μm in size. Dissolution and U–Pb analytical techniques were as outlined in Parrish et al. (1987), and error

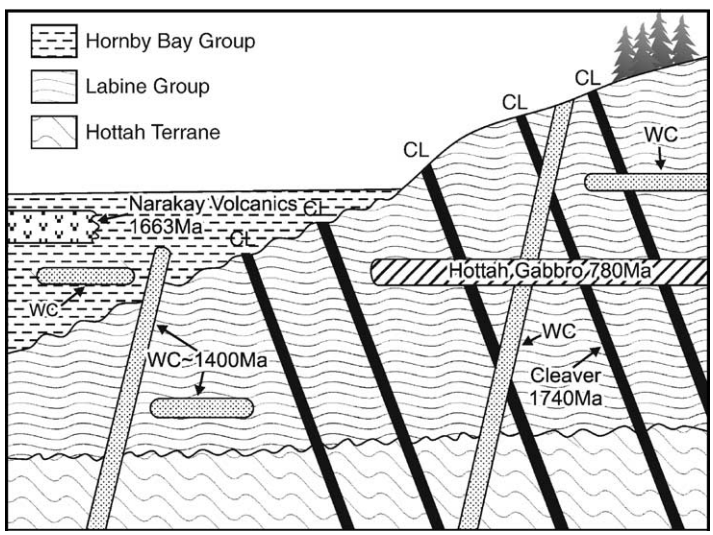


Fig. 3. Schematic relationships of Cleaver Dykes (CD, Fig. 2). Plutons of the Great Bear Arc are not delineated. CL: Cleaver Dykes. WC: Western Channel Diabase consists of both sills and dykes which are shown disconnected because they have small but significantly different directions of magnetization and may therefore have been intruded at slightly different times.

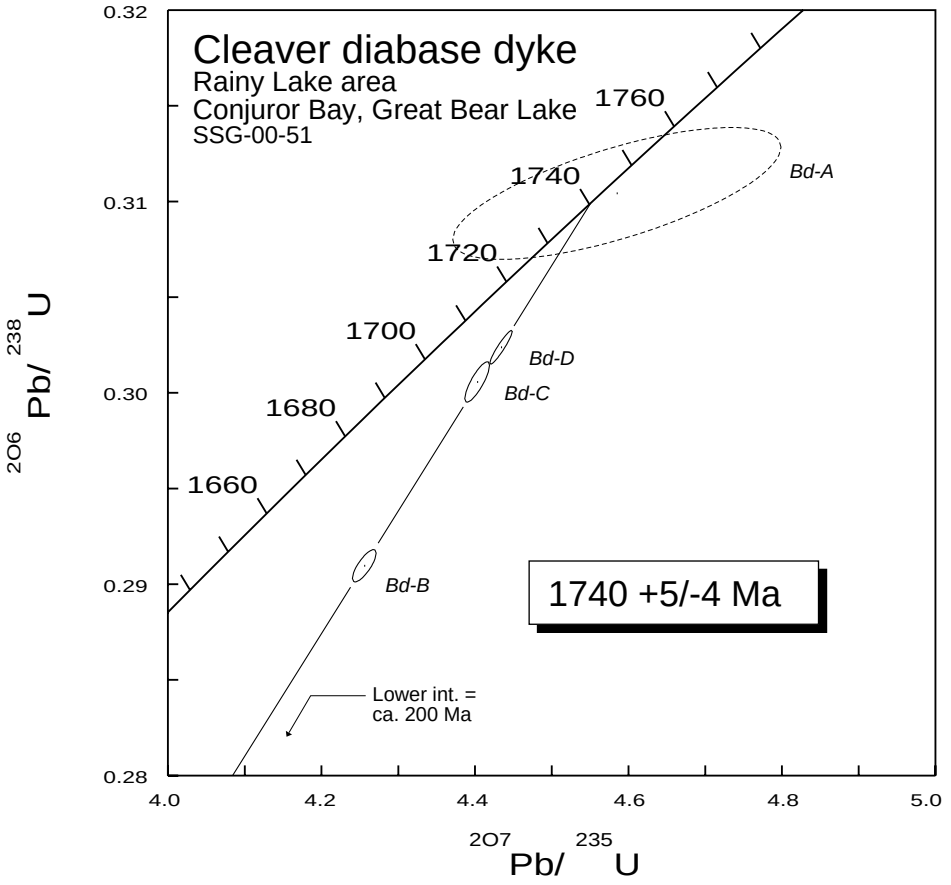


Fig. 4. Uranium–lead concordia diagram for baddeleyites from Cleaver Dyke. Upper intercept age reflects regression of all four points. Exclusion of the single imprecise fraction (dashed error ellipse, Bd-A) results in identical age and error. Error ellipses at 2σ level of uncertainty.

treatment as described by Roddick (1987). Results for four fractions are presented on a conventional concordia diagram in Fig. 4. Analyses range from concordant to 5.7% discordant, and define a co-linear array; a modified York (1969) regression analysis of all four fractions yields an upper-intercept age of 1740 $+5/-4$ Ma, and a lower intercept of approximately 200 Ma (MSWD = 0.2). We regard the former as a reliable estimate of the time of crystallization of the dyke, which, we suggest, applies to Cleaver Dykes as a whole, because their trend, general field characteristics, and, as we show below, their consistent magnetization directions, indicate they are a single coeval swarm.

3. Paleomagnetism of Cleaver Dykes

Oriented core samples (usually six per site) were obtained from 32 dykes. Most were sampled around Conjuror Bay in the southeast corner of Great Bear Lake over an area about 15 km × 25 km. One site, a resampling from an earlier study (site 172, Table 1), is on Echo Bay 50 km to the north. Two specimens were cut from each core to check reproducibility.

Ten dykes have magnetizations scattered about the present field which have low unblocking temperatures (T_{UB}) and remanent coercivities. They are probably Brunhes-age viscous remanent magnetizations (VRMs). Three dykes have very strong magnetization

Table 1
Directions of magnetization of Cleaver Dykes and associated rocks

Site	Latitude (°N)	Longitude (°W)	s	*n*	NRM (A/m)	Dec (°)	Inc (°)	*k*	α_{95} (°)
5	65.72	118.13	6	12	2.00	132.1	64.6	53	6.0
6	65.72	118.08	5	10	0.07	168.0	65.3	39	7.9
8	65.70	118.13	6	11	13.10	147.2	51.3	117	4.2
9	65.68	118.15	6	12	6.45	143.5	49.7	107	4.2
10	65.63	117.88	6	12	2.91	143.4	68.4	67	5.4
15	65.62	118.02	6	12	3.51	126.2	56.8	177	3.3
18	65.79	118.08	6	10	2.51	141.5	53.0	118	4.5
19	65.79	118.07	6	8	0.35	116.9	52.6	33	9.7
21	65.70	118.07	6	12	1.89	162.0	52.9	143	3.6
22	65.70	118.07	6	12	3.12	136.1	59.8	90	4.6
23	65.70	118.07	6	12	17.9	124.9	53.9	155	3.5
26	65.63	118.12	6	12	6.60	151.2	57.2	151	3.5
27	65.68	118.15	6	12	2.93	170.5	76.4	20	10.0
30	65.63	118.13	6	11	6.47	146.6	53.3	51	6.5
31	65.58	117.98	3	6	4.82	118.0	50.6	145	5.6
34	65.70	118.05	7	13	5.17	130.7	59.6	266	2.5
37	65.67	118.13	6	11	4.90	114.3	51.0	202	3.2
172	66.05	117.83	8	15	6.13	129.1	60.5	69	4.6
24	65.65	118.08	9	11	0.18	117.5	26.1	2	56.9
32	65.07	118.07	6	10	0.06	148.8	19.6	269	2.9
33	65.07	118.07	7	9	0.60	131.5	59.6	104	5.1
35	65.07	118.05	7	12	0.91	121.0	60.6	87	4.7
36	65.07	118.05	5	9	0.13	355.1	09.5	555	2.2
45	65.67	118.15	7	14	1.19	280.4	04.4	123	3.6

Notes. s(*n*), number of cores and specimens; NRM, intensity of natural remanent magnetization; Dec, Inc, declination and inclination with respect to horizontal; *k*, Fisher's estimate of precision; α_{95}, semi-angle of cone of confidence ($P = 0.05$). Site 24 is volcanic breccia (Fig. 9). Site 172 is resampling of site 144 of Irving et al. (1972a) at which Cleaver magnetization was first observed and which was mistakenly described by them as a Port Radium sill; this confirms their original determination *D*, $I = 138$ and 59° ($\alpha_{95} = \sim 9°$). Site 32 baked tuff at contact of dyke 22. Site 33 tuff at distance of 25 m from dyke 22. Site 35 baked andesite at contact of dyke 34. Site 36 andesite 50 m from dyke 34. Site 45 Gunbarrel Gabbro.

intensities, an order of magnitude greater than other dykes, which drop rapidly during alternating field demagnetization, and which decay linearly during thermal demagnetization; these are characteristics of lightning-induced isothermal remanent magnetization. Lightning is common in the area. Two dykes have no coherent magnetizations. These 15 dykes thus provide no useful information.

Seventeen dykes have stable magnetizations (Table 1). Thermal demagnetization curves are commonly square-shouldered, and T_{UB}'s are predominantly between 500 and 600 °C, typical of very fine grained magnetite (Fig. 5). Magnetizations are directed towards the southeast with intermediate downward inclination. We refer to this as Cleaver magnetization. It has "normal" polarity. Specimens with such ideal magnetization were demagnetized in one step. If appreciable Brunhes VRMs were present, specimens were demagnetized in three to five steps, as required to remove it. Table 1 and Fig. 6a give results by sites. These are not recent data, and it would be usual nowadays to demagnetize all at numerous steps. That would have made no difference; the mean direction of 17 sites obtained by line-fitting to data from multiple demagnetization steps from the 34 pilot specimens is statistically identical to that obtained by line-fitting data from 126 specimens demagnetized at three to five steps and single step demagnetization in 53 specimens (Fig. 6b). We conclude that the omission of multiple demagnetization steps in some specimens (53 out of a total of 213) has not biased the result.

In the neighborhood of the Cleaver Dykes are two other intrusive diabase suites (Fig. 3), the approximately 1400 Ma (Wanless et al., 1970; Wanless and Loveridge, 1978), Western Channel Diabase (Irving et al., 1972a; Hildebrand, 1983b), and

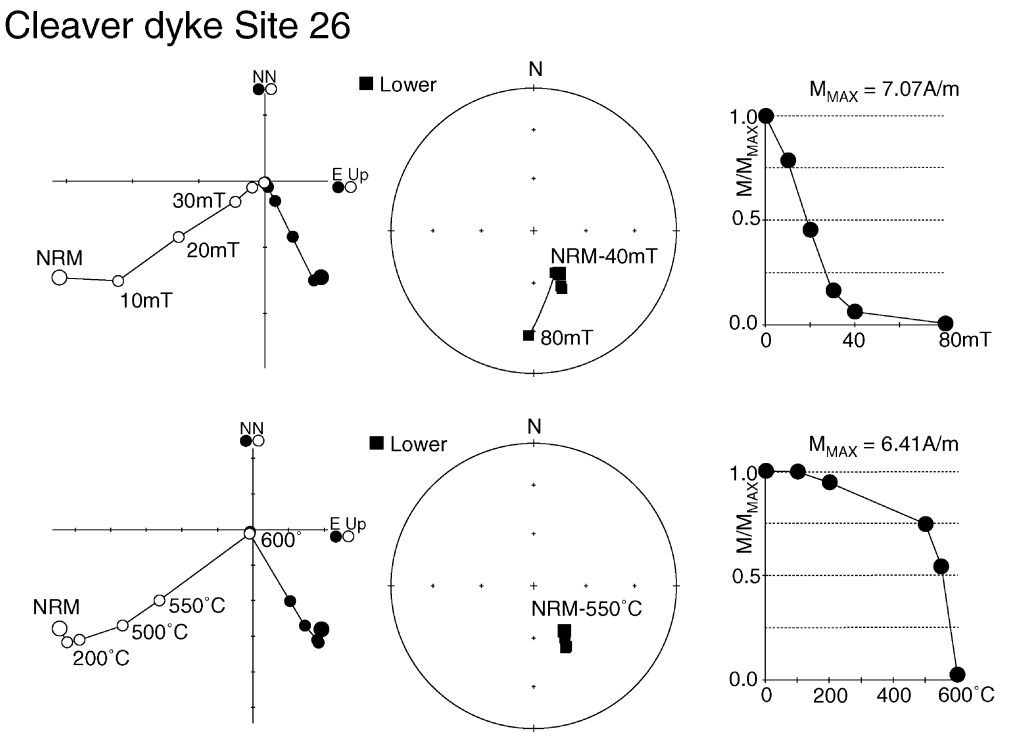


Fig. 5. Cleaver magnetization at site 26 relative to present horizontal. Left, orthogonal demagnetization plots on N-E and N-upward planes. Centre, equal-area plots of directions all down. Right, magnetization intensity vs. alternating field (AF) strength (above), and temperature (below). NRM is natural (initial) remanent magnetization.

the approximately 780 Ma Hottah Gabbro sheets (LeCheminant and Heaman, 1994). Extensive contact tests showed that both have magnetizations acquired at the time of cooling following emplacement. The Cleaver Dyke at site 172 is cut by a sill of the Western Channel Diabase which has been sampled at five sites at distances between 0 and 4 km from the dyke. A Hottah sheet (called the Gunbarrel Gabbro) has been sampled at two sites (site 45 Table 1, and site 1 in Park et al., 1995). In both instances, the directions in later intrusions are very different from Cleaver magnetization (Fig. 6a), which therefore pre-dates both intrusive suites and has not been influenced by them.

Directions in baked andesite at the contact of dyke 34 are in excellent agreement with the dyke itself, while andesite 50 m distant directions are very different (Fig. 7b). The corresponding thermal demagnetization diagrams are given in Fig. 8. The diabase has a stable single component magnetization with "magnetite" T_{UB} of 500–550 °C characteristic of other Cleaver Dykes (Fig. 5). Andesite at the contact has the same magnetization direction with "magnetite" and "hematite" T_{UB}'s 500–550 and 600–700 °C, respectively. Andesite at a distance has similar demagnetization characteristics but an entirely different direction. At another dyke (22), baked tuffaceous rock at the contact is in excellent agreement with the dyke itself, whereas similar tuffaceous rock 25 m away has a very different direction (Table 1, Fig. 7a). Intensities of magnetization of contact rocks are 4–10 times greater than at a distance (Table 1).

As a further check, several cores were taken from three large andesite boulders (site 24) in a volcanic breccia within the Labine Group. Dispersion within boulders is variable and generally small. Between boulders it is large, and all directions are very different from Cleaver magnetization (Fig. 9). The boulders are older than Cleaver diabase, but their dispersed directions show that, since they were incorporated in the breccia, there has been no general remagnetization

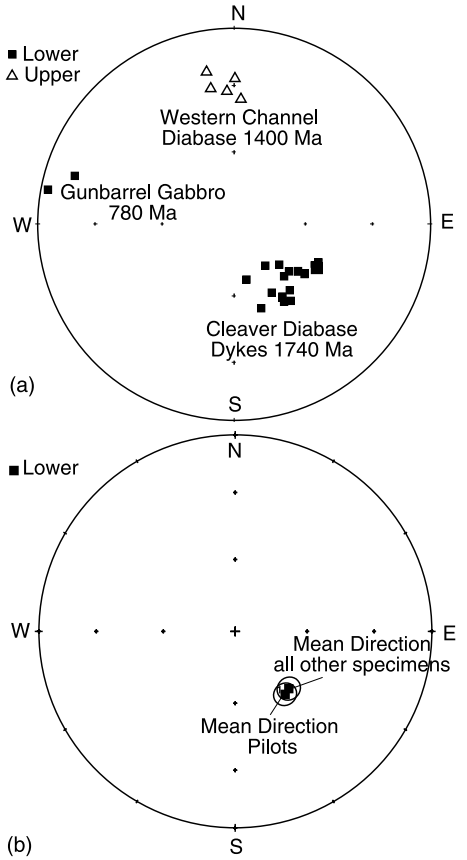


Fig. 6. (a) Magnetization directions at each sampling site. Also plotted are directions in Hottah sheet (Gunbarrel Gabbro) at two sites (site 45 Table 1, and a site described by Park et al., 1995), and at five sites in a sill of Western Channel Diabase immediately north of site 172 (E. Irving, unpublished data). (b) Mean directions from pilot specimens from 17 dykes (140.7 and 56.6°, $k = 65$, $\alpha_{95} = 4.7°$, $N = 17$) and from other specimens (135.2 and 57.5°, $k = 58$, $\alpha_{95} = 4.7°$, $N = 17$) compared. All directions relative to present horizontal. These results do not differ from one another or from the mean using all specimens as given in Table 2.

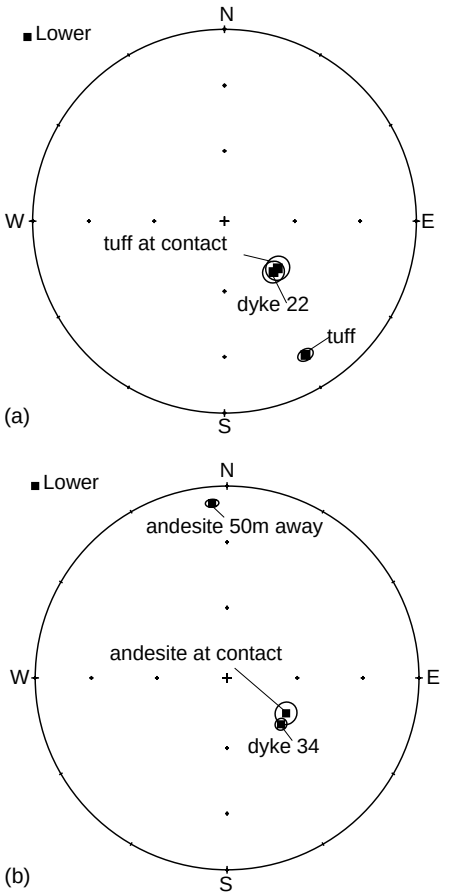


Fig. 7. Direction of magnetization in Cleaver diabase at sites 22 (a) and 34 (b) and in nearby rocks of the Labine Group. Errors: $P = 0.05$. All directions relative to present horizontal.

episode such as might have affected Cleaver Dykes. These three tests, especially the contact test of Figs. 7 and 8, together with characteristically high T_{UB} and the excellent grouping of directions (Table 2, Fig. 6a), show that Cleaver magnetization was acquired during initial cooling soon after intrusion.

Table 2
Mean directions of magnetization

	B	Dec (°)	Inc (°)	*k*	α_{95} (°)	Pole (°N, °W, *K*, A_{95})
(1) Hornby Sandstone (horizontal)	4	134.0	40.8	61	11.8	6.3, 76.5, 53, 12.7
(2) Hornby Sandstone (bedding)	4	39.5	42.4	42	14.3	6.3, 82.5, 36, 15.6
(3) Cleaver Dykes (horizontal)	17	136.5	57.4	64	4.5	19.4, 83.3, 36, 6.1

Notes. *B* is the number of collecting sites. Paleopoles calculated as averages of site poles, *K* their precision, A_{95} their error. Other symbols as in Table 1.

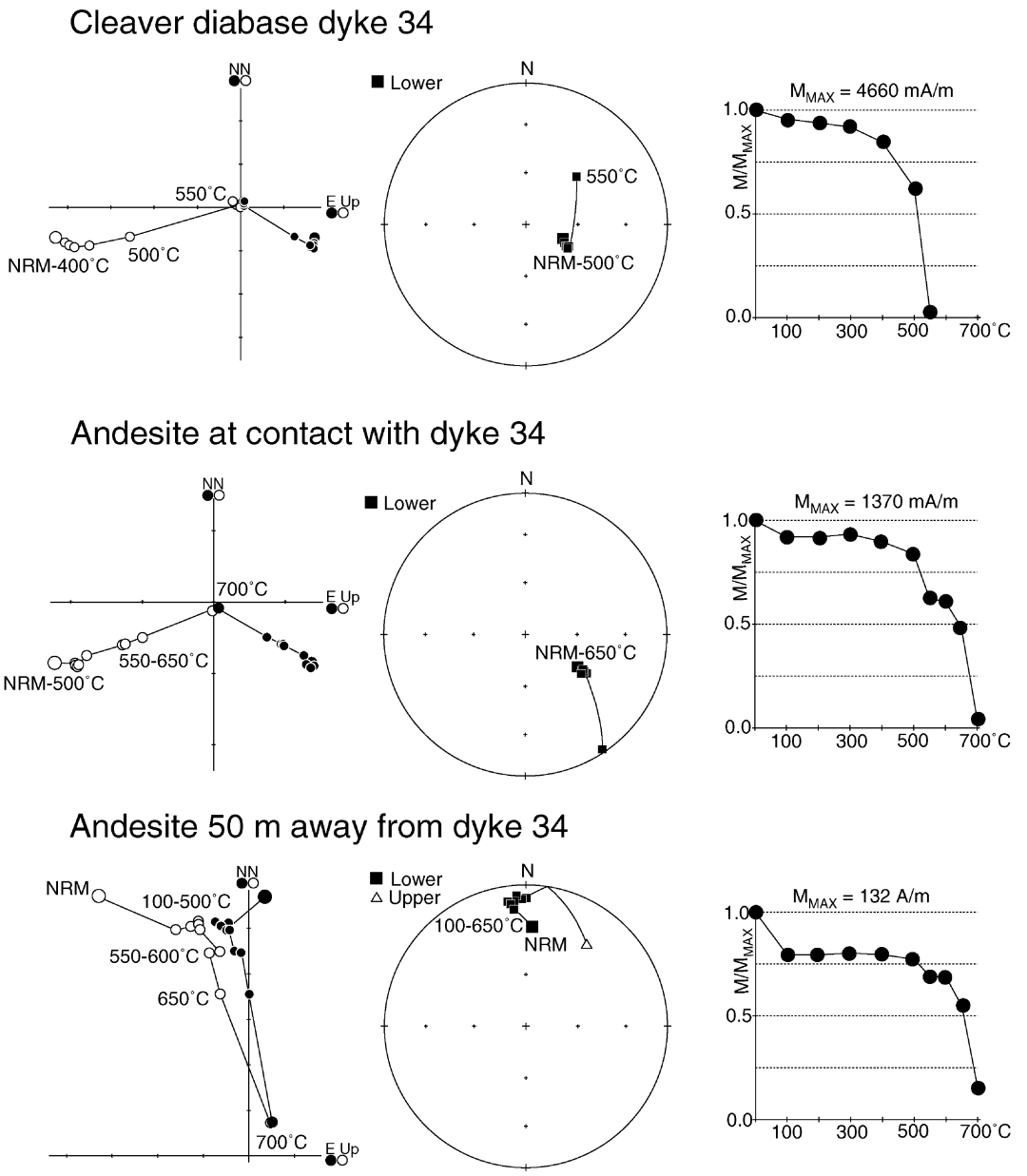


Fig. 8. Thermal demagnetization of Cleaver diabase (above) and andesite of the Labine Group at contact (center) and at a distance (below). Conventions in Fig. 5.

4. Paleomagnetism of Hornby Bay Sandstone

Sandstones of the Hornby Bay Group are mainly grey-colored. Towards the base there are restricted red strata, the result of deposition of hematite during early diagenesis (Kurt Kyser, personal communication, 2002). Four stratigraphic levels spanning 47 m have been sampled from the shore of Hornby Bay on Great Bear Lake. The locality is marked A56 on Fig. 2 of Irving et al. (1972a,b). Thermal demagnetization

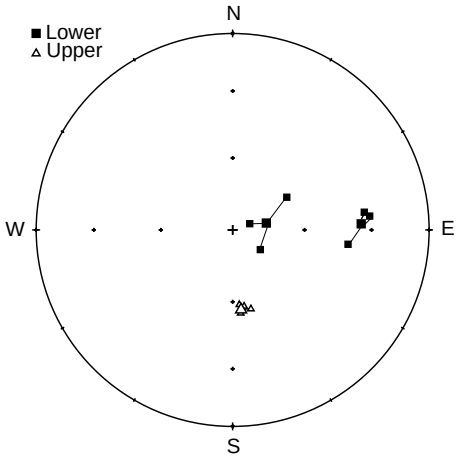


Fig. 9. Magnetization directions of specimens from three andesite boulders in a volcanic breccia of the Labine Group. Means have larger symbol. Relative to present horizontal.

(Fig. 10) yields square-shouldered curves with T_{UB}'s mainly between 650 and 700 °C, characteristic of hematite and directions confirming those reported earlier by Irving et al. (1972a). Directions are generally similar to Cleaver magnetization but inclinations are shallower (Table 2).

5. Post-Hudsonian paleopoles from western Laurentia

It has long been recognized (see, e.g. Evans et al., 1980; Park, 1975) that many secondary (or overprint) magnetizations in Early Proterozoic sedimentary rocks and in rocks that have undergone high-grade metamorphism in the Hudsonian Orogeny, or its equivalent (Nagssugtoqidian Orogeny) in Greenland, give paleopoles clustered in the Central American region (Fig. 11). All have "normal" polarity. We now consider this Central American cluster of paleopoles and the magnetizations from which they were derived. We refer to them collectively as post-Hudsonian paleopoles and magnetizations. It is important to note that post-Hudsonian magnetizations occur across Laurentia, but despite extensive study have not been observed in the Grenville Province. Consequently, they are likely to have been acquired after the assembly of all major units of Laurentia and before the accretion of the Grenville Province at 1200–1000 Ma (Hoffman, 1989).

Post-Hudsonian magnetizations have three different origins. First are nine overprints observed in bedded sequences—six are post-tilting magnetizations from the Great Slave Supergroup (G1, G2, TO2), the Cape Smith Belt (CS), and the Belcher and Ottawa Islands in Hudson Bay (EP, KO), two are early diagenetic magnetizations (AA, HS), and one is a post-depositional magnetization in conglomerate clasts (MC) in the Hudsonian Orogen (Fig. 11). The post-tilting magnetizations are well grouped before but scattered after correction for tilt of bedding, have T_{UB}'s of 300–500 °C, and have direction far removed from the present field, so are not Brunhes VRMs. The second group comprise seven occurrences of single-component magnetizations acquired during uplift and

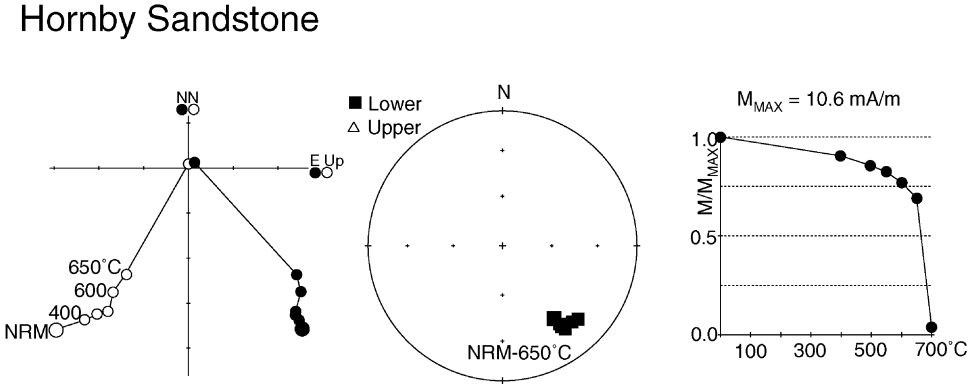


Fig. 10. Thermal magnetization of Hornby Bay Sandstone. Conventions in Fig. 5.

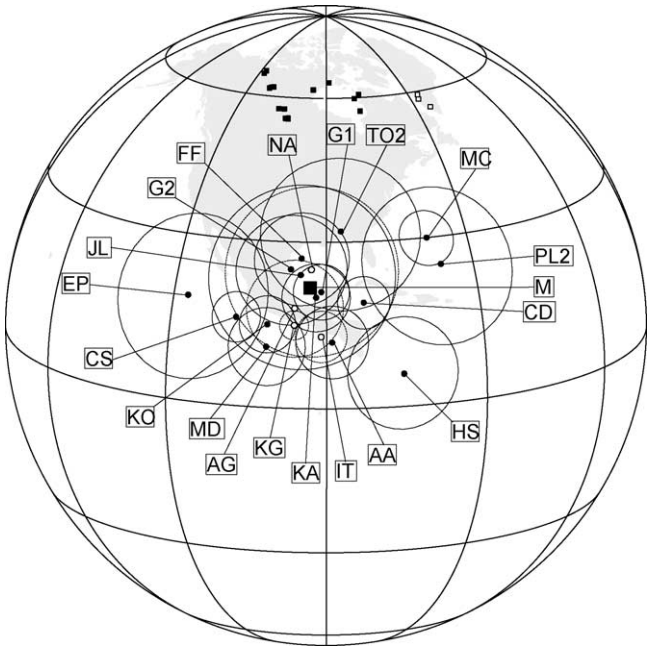


Fig. 11. Post-Hudsonian paleopoles from Laurentia, including Greenland. The large square (■) M is their mean. All paleopoles calculated with rocks in their present position, errors at $P = 0.05$. Solid symbols from Canada, open symbols from Greenland. All magnetizations are "normal", N-seeking directions point towards the paleopoles plotted. Greenland paleopoles rotated -11° about 71.5°N, 126.3°W. In the following notes "author(s) refers to workers cited. AA: first diagenetic magnetization in Athabasca Sandstone about 1650–1700 Ma (Kotzer et al., 1992). AG: Amitsoq Gneiss Greenland (Fahrig and Bridgwater, 1976); magnetization parallel to KG; authors consider age of both to be 2000–1800 Ma; for reasons given under KG we consider them younger. CD: Cleaver Dykes, herein. G1 and G2 post-folding (Hudsonian Orogeny) magnetization in red beds of Kahocella Group, GSS (Evans et al., 1980; Irving and McGlynn, 1979). CS: Cape Smith basalt, magnetization post-dates Hudsonian deformation (Fujiwara and Schwarz, 1975). EP: Eskimo Point volcanics, magnetizations post-date Hudsonian Orogeny and is possibly related to Rb–Sr isochron 1713 ± 25 Ma (Schmidt, 1980). FF: Flin-Flon–Snow Lake greenstones; "A" magnetization of Park (1975) who suggested an age of 1700–1600 Ma; Park also compiled K–Ar mica ages for sampling area which averaged 1760 Ma in good agreement with both age and paleopole of Jan Lake Granite. HS: Hornby Bay Sandstone, herein. IT: Itivdelq Dykes, Greenland (Morgan, 1976), author relates age of magnetization to "setting of K–Ar mineral clocks " which in the area range 1790–1650 Ma; second study of these dykes (Beckman and Mitchell, 1976) yields essentially the same result and authors suggest magnetization acquired by cooling and uplift at "about 1750 Ma" based on K–Ar data. JL: Jan Lake Granite (Gala et al., 1995), 1759 +5/−4 Ma (Bickford, personal communication, see text). KA: metamorphosed Kaminak Dykes, acquired after metamorphism, magnetization, possibly related to K–Ar ages which range 1812–1615 Ma (Christie et al., 1975). KG: Kangâmiut metadiabase dykes, Greenland (Fahrig and Bridgwater, 1976); these and the Amitsoq Gneisses into which they are intruded have paleopoles, like Cleaver and Jan Lake paleopoles; $^{40}Ar/^{39}Ar$ studies yield plateau ages ranging 2028–1697 Ma (Willingers et al., 1999); 2 Ga ages agree well with a U–Pb zircon age and presumably record dyke emplacement; five younger plateau ages range 1756–1697 Ma in approximate agreement with Cleaver and Jan Lake dates, and therefore we suggest they reflect age of magnetization acquired upon uplift and cooling. KO: komatiites, Ottawa Islands, post-Hudsonian (Buchan and Baragar, 1985). MC: Missi conglomerate clasts, post-depositional pyrrhotite magnetization (site 2 of Symons and Lewchuk, 1994). MD: amphibolites of Melville-Daly Bay (Park, 1973); mica K–Ar ages range from 1690 to 1580 Ma substantially younger than age of magnetization implied by this paleopole. NA: Nagssug Gneisses metamorphosed and deformed in Nagssugtoquidian Orogeny (Beckman et al., 1977); magnetization acquired during uplift and cooling, which the authors imply, was between 1850 and 1600 Ma. PL2: Peter Lake Domain, overprints whose age is estimated to be 1780 ± 30 Ma by Symons (1994). TO2: Tochatwi Formation, post-folding overprint in red beds (Evans and Bingham, 1976).

cooling following moderate to high-grade metamorphism (AG, FF, IT, KA, KG, MD, NA); in these rocks older magnetizations have been destroyed by recrystallization of the rocks during metamorphism. K–Ar ages and some Rb–Sr isochrons from these bodies generally range in age 1800–1600 Ma (see legend to Fig. 9), and Morgan (1976) for instance has suggested that the magnetizations are of this age

(IT). Third, there are overprints of moderate stability in Archean and Early Proterozoic igneous rocks that long post-date their emplacement (PL2). Older magnetizations are commonly present, in the first and last groups (e.g. KA and SE of the next section).

Among post-Hudsonian paleopoles, there are two from precisely dated unmetamorphosed intrusive rocks. These are from the 1740 Ma Cleaver Dykes (CD) just described and the Jan Lake Granite (JL). The latter is a suite of post-orogenic granite, pegmatite and aplite intrusions of the Hanson Lake Block, in the Hudsonian Orogen (Fig. 2). It has yielded a Rb–Sr whole-rock isochron age of 1750 ± 20 Ma (Bell and MacDonald, 1982) and a U–Pb zircon age of 1759 ± 1 Ma (M.E. Bickford, personal communication, 2002). Gala et al. (1995) recognized two magnetizations in the Jan Lake Granite, "A" directed to the SE with positive inclination, and "B" directed steeply downward. They regard "A" as "probably older that 1844 Ma", on the grounds that the complex also contains older rocks which may have inherited an older magnetization. They regard "B" as acquired "probably about 1770 Ma". We agree that "A" pre-dates "B", but suggest that both are younger than Gala and colleagues proposed, for these reasons. "B" magnetization is very close in direction to the youngest diagenetic overprint in the nearby Athabasca Sandstone dated at about 900 Ma by Kotzer et al. (1992), and could therefore be a Neoproterozoic overprint. "A" is not significantly different from Cleaver magnetization and has the same polarity. "A" is based, in part, on some excellent square-shouldered demagnetization curves with T_{UB}'s about 550 °C. We suggest that it, like Cleaver magnetization, is most likely a TRM acquired at the time of intrusion, namely, approximately 1759 Ma. The Jan Lake and the significantly younger Cleaver paleopoles are in good agreement and fall in the cluster of post-Hudsonian paleopoles (Fig. 11). We therefore suggest that because of their similar magnetization directions and polarity, post-Hudsonian magnetizations were acquired close to the same interval in which these two bodies were intruded. We recognize that many of these secondary magnetizations do not have strong stratigraphic and geochronometric age constraints (details in legend to Fig. 11), but none are inconsistent with our suggestion.

Much of the dispersion of post-Hudsonian paleopoles is attributable to statistical errors (Fig. 11). Individual paleopoles may also have been affected by small local tilts and rotations (EP for instance), and some are based on limited sampling (AA, HS, MS). Such effects, if present, must not be large, otherwise there would be no grouping, and taken together their effects are probably random. Their mean giving equal weight to each paleopole is 20.2°N, 92.3°W, $K = 42$, $A_{95} = 5.7°$ ($N = 19$). Omitting AA, MS, and HS because of limited sampling, the mean becomes 20.8°N, 94.5°W, $K = 51$, $A_{95} = 5.2°$ ($N = 16$), which is our preferred estimate of the mean geomagnetic field during the interval approximately 1740–1760 Ma.

6. Syn-Hudsonian paleopoles from western Laurentia

There are 18 paleopoles from syn-Hudsonian rocks of western Laurentia (Table 3, Fig. 12). It is important to note that we only discuss here data from western Laurentia as defined in Section 1. Sixteen paleopoles are derived from rock units that contain reversals of magnetization, directions being to the northwest ("reversed") or southeast ("normal") with inclinations shallower than post-Hudsonian magnetizations. Fifteen are from red beds and igneous strata which occur in four very thick successions, the Coronation, Dubawnt, Goulburn and Great Slave Supergroups (Fig. 2). These strata were deposited, folded and faulted (but not metamorphosed) during the Hudsonian Orogeny. In each study, the dispersion of magnetization directions decreases after correction for tilting so they were acquired before deformation. Most (16) have been calculated after restoration of directions to bedding. Two have been calculated from directions relative to present horizontal—the precisely dated sub-vertical Sparrow Dykes (SD), and the voluminous Wathaman Batholith (WA) 500 km to the southeast, whose paleopoles are in excellent agreement (SD and WA of Fig. 12), so neither body is likely to have been substantially tilted, rotated or moved laterally relative to the other since acquiring their magnetizations. There is a result from Peter Lake Domain (PL1) just north of the Trans-Hudson suture which was probably magnetized upon cooling after high-grade metamorphism. Finally, there is a result from the Et-Then Group red beds that unconformably overlie the Great Slave Supergroup (ET).

Table 3
Syn-Hudsonian rocks from western Laurentia, age approximately 1835 Ma

Rock unit (reference)	*B*	% "rev"	RR	RPD	Notes
AK Akaitcho River Fm. (6)	35	74	cw26 ± 11	10 ± 10	Redbeds, GSS
DP Douglas Peninsular Fm. (8)	6	33	cw33 ± 15	10 ± 14	Redbeds, GSS
BL Baker Lake Group (11)	30	67	–	–	Redbeds, volcs, DS
KA Kahocella Group (12)	18	50	–	–	Redbeds, GSS
MA Martin Fm. (3)	15	66	–	–	Redbeds, andesites
MR Mara Fm (5)	25	85	cw37 ± 11	1 ± 10	Redbeds, GS
PE Pearson Fm. (7)	12	100	–	–	Basalt, GSS
PH Peacock Hill Fm. (5)	15	73	cw21 ± 12	10 ± 11	Redbeds, GS
ET Et-Then Group (9)	14	71	ccw23 ± 11	9 ± 10	Redbeds.
PS Peninsular Sill (8)	7	86	cw29 ± 11	16 ± 10	Gabbro, CS
PL1 Peter Lake Domain (13)	15	100	ccw22 ± 12	4 ± 11	Metamorphic rocks
SE Seton Fm. (8)	19	0	cw22 ± 11	8 ± 9	Redbeds, lavas, GSS
SP Sparrow Dykes (10) (2)	10	10	–	–	Diabase
ST Stark Fm. (1)	39	51	cw76 ± 10	12 ± 9	Redbeds, GSS
TA Takiyuak Fm (8)	17	12	cw42 ± 11	6 ± 10	Redbeds, CS
TO Tochatwi (4)	8	50	cw74 ± 13	15 ± 12	Redbeds, GSS
WA Wathaman Batholith (14)	19	0	–	–	Monzonite
WR Western River Fm. (5)	22	23	ccw57 ± 10	5 ± 11	Redbeds, GS

References: (1) Bingham and Evans (1976), (2) Bostock and van Breeman (1992), (3) Evans and Bingham (1973), (4) Evans and Bingham (1976), (5) Evans and Hoye (1981), (6) Evans et al. (1980), (7) Irving and McGlynn (1978), (8) Irving and McGlynn (1979), (9) Irving et al. (1972b), (10) McGlynn et al. (1974), (11) Park et al. (1973), (12) Reid et al. (1982), (13) Symons (1994), (14) Symons (1991).
Notes. *B*, number of sites, several samples at each; % "rev", percent "reverse" polarity; RR, relative rotation; cw, clockwise; ccw, counter-clockwise; RPD, relative paleolatitudinal displacement; both relative to reference field given in Section 6; see also Fig. 2. CS: Coronation Supergroup; DS: Dubawnt Supergroup; GS: Goulburn Supergroup; GSS: Great Slave Lake Supergroup.
Ages: BL 1850–1785 Ma (Rainbird et al., 2001). MA 1835 ± 50, K–Ar on biotite from mafic dyke presumed feeder to Martin Fm. (Frazer et al., 1970). PE 1809 ± 19 Ma, Rb–Sr isochron and must be considered to be "magmatic" age (Goff et al., 1982). PL magnetization "probably acquired during regional metamorphism" at 1869 ± 29 Ma. PS incomplete contact test indicates magnetization acquired during cooling following intrusion. SE 1804 ± 23 Ma, Pb/Pb on "basalt" with reasonable agreement with earlier Rb–Sr dates of 1833 ± 28 Ma and 1805 ± 16 Ma on same rocks (Cummings, 1980); the Compton laccoliths which intrude many units of the Great Slave Supergroup and which postdate the deformation of the latter have zircon U–Pb ages of 1865 ± 15 Ma (Bowring et al., 1984), so the Seton Formation, and other associated units of the Great Slave Supergroup, the Akaitcho Formation (AK), for example, probably are substantially older than the two Rb–Sr dates. SP 1827 ± 4 Ma, U–Pb baddeleyite age (Bostock and van Breeman, 1992). WA 1854 ± 11 Ma, mean of six U–Pb zircon dates summarized in Symons (1991). WR The portion of the Western River Formation sampled paleomagnetically and described in Ref. (5) is now called the Rifle Formation and is dated (U–Pb on zircons from ash beds) at 1963 ± 6 Ma (Bowring and Grotzinger, 1992).

The paleopoles are spread over a huge arc centered about western Laurentia, the hallmark of large differential rotations of sampling localities about local vertical axes (Fig. 12). The paleolatitudinal (essentially the present north-south) spread is about 20°, for which there are three possible reasons. First, much but not all, is attributable to statistical errors. Second, there are uncertainties due to difficulties in estimating paleohorizontal in stratified rocks, especially in lavas (e.g. PE and lavas in BL), which are likely to be random over the wide range of rocks studied. Third, there are uncertainties from incomplete removal of overprints, which we now consider.

As we have just described, the widespread overprinting that followed the Hudsonian Orogeny always has the same "normal" polarity (Fig. 11). It has been well studied in several syn-Hudsonian sequences, in which its effects have been largely removed by thermal demagnetization. However, much of the data are over 20 years old, and it is possible that some overlaps in the spread of unblocking temperatures of primary and overprint magnetizations may not have been identified; small overprints may remain. If they do, they would increase (decrease) estimates of inclination (and paleolatitude) of southerly ("normal") directed magnetizations, and decrease the inclination

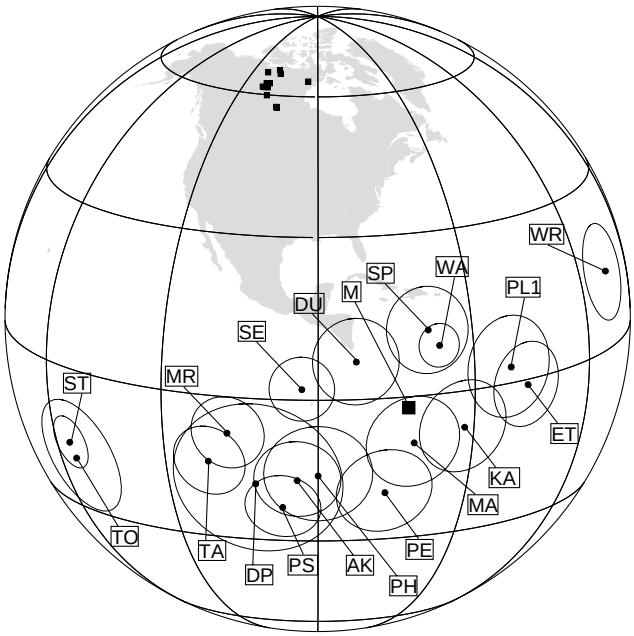


Fig. 12. Syn-Hudsonian paleopoles for western Laurentia. Magnetization observed at localities indicated point to the 18 paleopoles shown (if normal polarity) or to their antipole (if reversed). Paleopoles labeled as in Table 3. Errors: $P = 0.05$. The square (■) labeled M is reference paleopole for western Laurentia. Direction of magnetization all corrected for local tilt, except SP and WA which are intrusives and PL 1a metamorphic complex.

(and paleolatitude) of northerly ("reversed") directed magnetizations (Reid et al., 1982). Mean directions from which the paleopoles of Fig. 12 have been obtained have been calculated by reversing "reversed" magnetizations, thereby inverting any residual overprints, which are thereby cancelled out, provided "reversed" and "normal" magnetizations are about equal in number. They are not equal in individual results (Table 3), but as we show below, they are approximately equal when all are considered together. Hence, results with a preponderance of "reversed" ("normal") magnetization should yield paleopoles displaced to the south (north). Consistent with this are the predominantly normally magnetized Sparrow Dykes [10% "rev"] and their northerly paleopole, SD. The Seton Formation paleopole SE, which is derived entirely from normal magnetizations, is $18 \pm 8^{\circ}$ north of that for the Akaitcho River Formation (AK), its approximate stratigraphic equivalent, which also is consistent. Other consistent examples are the Peninsular Sill PS [94% "rev"], the Pearson Formation PE [100% "rev"] and the Peacock Hill Formation PH [73% "rev"] which are all predominantly reversed and have the predicted southerly paleopoles. By contrast the Baker Lake Group is predominantly "reversed" [67%] and yet its paleopole is northerly. No effect is evident in the Mara MR [86% "rev"] and Takiyuak [15% "rev"] formations which have opposite polarity bias and yet are in mutual agreement, which we interpret to mean that syn- and post-Hudsonian magnetizations in them have been cleanly separated. We conclude that some paleopoles probably are affected by small residual overprints and others are not, but that there is unlikely to be any systematic effect on the mean trend of the arc of Fig. 12 because, as we show below, the total numbers of "normal" and "reversed" magnetizations are evenly matched.

The excellent agreement between paleopoles from the Sparrow Dykes and the Wathaman Batholith has already been noted, indicating that it is unlikely that either of them has been substantially tilted, rotated or moved laterally relative to the other. They, and the Baker Lake, Martin, Pearson, Kahocella bedded sequences, have similar declinations, indicating that they too have not been rotated very much relative to one another. We suggest therefore that the reference

paleopole for western Laurentia may be estimated by forming the average of these six results, which is 1.1°S, 71.7°W, $A_{95} = 12.1°$. Rotations (RRs) and paleolatitudinal displacements (RPDs) of other bodies of Fig. 12 relative to this reference paleopole are listed in Table 3 and plotted in insets in Fig. 2. We first consider relative rotations

Western Laurentia is laced with large transcurrent faults, which Gibb (1983) has suggested were caused by the essentially simultaneous indentation of the Slave and Superior cratons into the Rae and Hearne Provinces (Fig. 13). There are several phases of compression with changes in direction of motion (Hoffman, 1989). Particularly notable are the >70° clockwise rotation of the Stark and Tochatwi Formations, presumably reflecting dextral shear along the Great Slave Shear Zone that passes just south of the sampling area (GSSZ of Fig. 2). Further north, there are smaller clockwise rotations in the Coronation Supergroup, perhaps reflecting the dextral-oblique subduction invoked by Hoffman (1982, 1989). The large counter-clockwise rotation of the Western River Formation at the base of the Goulburn Supergroup and the oldest of the syn-Hudsonian sequences studied (Bowring and Grotzinger, 1992, Table 3 legend), may reflect early sinistral shear along the Bathurst Fault (BFF, Fig. 2). The magnetization directions in the younger Mara Formation indicate a small clockwise rotation, reflecting perhaps a later motion of opposite sense.

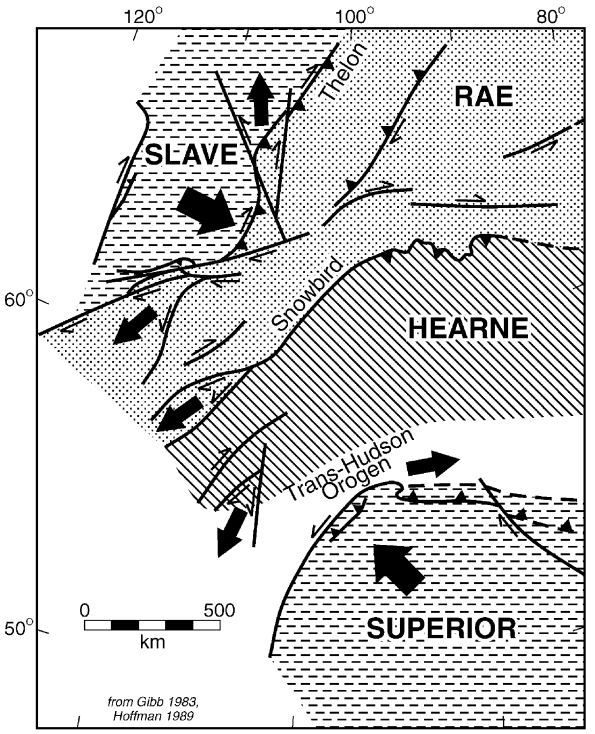


Fig. 13. Extension of crustal blocks and their bordering transcurrent faults caused by indentation of Slave and Superior cratons as proposed by Gibb (1983) and modified by Hoffman (1989).

Relative latitudinal displacements are either small or not statistically significant (Table 3), and, as we have just shown, these small divergences could reflect incomplete removal of post-Hudsonian overprints. This implies that, to a good approximation, the various elements of western Laurentia have remained at about the same paleolatitude during the interval in which the syn-Hudsonian rocks of Table 3 were formed; there is certainly no evidence presently available that their latitude changed greatly. The oldest is the Western River Formation (1963 ± 6 Ma) and the youngest the Sparrow Dykes (1827 ± 4), a range well in excess of 100 million years The magnetization of the Western River Formation could be secondary, acquired long after deposition, but the occurrence in it of reversals would indicate otherwise.

The Et-Then Group red beds has pre-tilting magnetizations and both “normal” and “reverse” polarities. Because it overlies unconformably the Great Slave Supergroup, it has been customary to consider that their magnetizations was acquired at some significantly later time than that of the Great Slave Supergroup (Reid et al., 1982; Irving and McGlynn, 1981). However, there is now no good reason for this, its paleopole falls on the same general arc as those from other syn-Hudsonian sequences, and its paleolatitude is similar (Fig. 12), so its age could fall in the above wide range. That blocks of continental crust should have remained little changed in latitude over such long intervals is not unprecedented—Antarctica since the Early Cretaceous, for example (Smith et al., 2000).

7. Paleolatitudes and paleomeridians

Figs. 14 and 15 show paleolatitudes and paleomeridians for western Laurentia. For the time represented by syn-Hudsonian sequences (ca. 1960–1830 Ma),

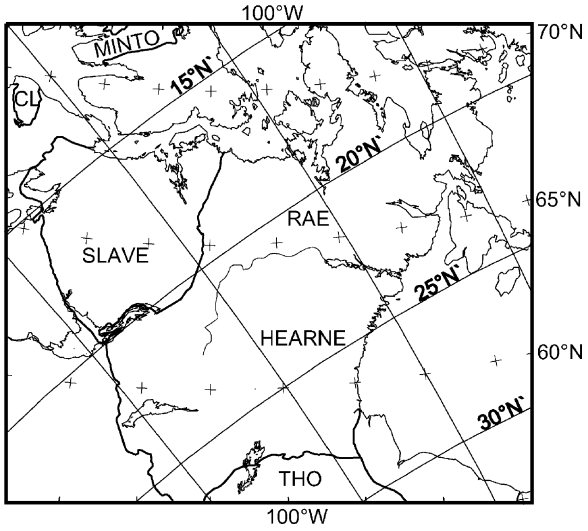


Fig. 14. Paleolatitude map for range approximately 1960–1830 Ma. Present latitude and longitude are given around the margin. Paleolatitudes are designated N′ consistent with the polarity convention explained in Section 1; it is not determined whether western Laurentia was in the northern or southern hemisphere.

during which the Coronation, Goulburn, Dubawnt, Great Slave Supergroups and Et-Then Group were deposited, paleolatitudes are 15–30°, and paleomeridians trend northwest-southeast relative to the present north. For post-Hudsonian time (ca. 1760 Ma), during which deposits of the Thelon, Athabasca, and Amundsen Basins were laid down, paleolatitudes range from 35 to 50° and paleomeridians trend northnorthwest-southsoutheast relative to the present north. Together with the abundance of climatically sensitive sedimentary rocks for these intervals, the maps of Figs. 14 and 15 now make possible a detailed study of Early Proterozoic paleoclimates of the western Shield. One general point is that Fig. 14 is consistent with the widespread occurrence in the western Shield of carbonate strata, arid climate deposits, such as playa-lake deposits, evaporites and redbeds at that time, whereas there is no comparable widespread occurrence of them in post-Hudsonian deposits consistent with their higher paleomagnetically determined latitude.

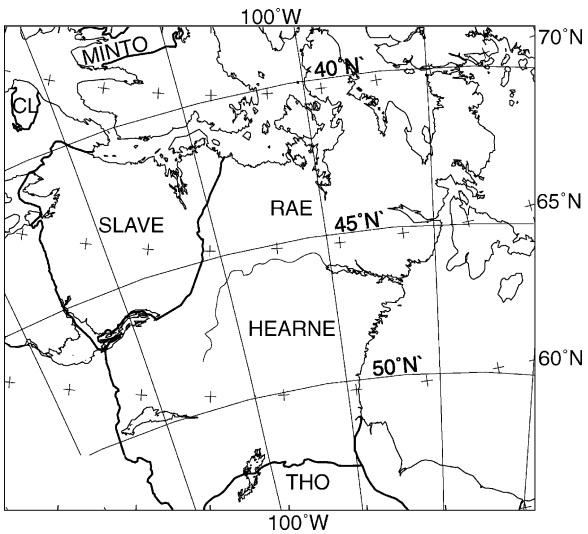


Fig. 15. Paleolatitude map at approximately 1760–1740 Ma. See Fig. 14 for conventions.

If our arguments, that the latitudes of western Laurentia change little within each of these two intervals, and if our estimates of their substantial durations are correct (ca. 1960–1830 and 1760–1740 Ma), then the way is now clear to making quantitative comparisons at these times with Precambrian cratons world-wide; these intervals of latitudinal stability are certainly long enough to make this project geochronologically feasible.

8. Reversals

As already noted all post-Hudsonian magnetizations represented in Fig. 11 have “normal” polarity, they occur across the Laurentia from Great Bear Lake to Greenland (Fig. 1). Some were acquired by hematite as it was precipitated from basinal fluids during burial diagenesis (AA; HS). Others are carried by magnetite and were probably acquired as a result of cooling during uplift following metamorphism (e.g. KA, KG, MD, NA). Yet others (CD, JL) probably were acquired during cooling following igneous crystallization. These different processes may have operated at the same time somewhere across Laurentia, but it is highly unlikely that the record of them, obtained by such a haphazard sampling of such diverse rocks and settings, would span only a brief interval. Also, the dates of the Jan Lake Granite and the Cleaver Dykes are significantly different, but their paleopoles and polarity are in agreement. Therefore, we argue that the post-Hudsonian magnetizations of Fig. 11 were acquired over an

interval of at least 13 million years (the limits of dating errors) during an interval towards the end of the Early Proterozoic when the geomagnetic field had constant, "normal" polarity; the interval could have been much longer. We propose the name Cleaver Superchron for this interval, because it is characterized by the geomagnetic field direction carried by the Cleaver Dykes whose age is accurately determined. Historically, the name is appropriate because it was from what is now recognized as a member of this dyke swarm that the post-Hudsonian magnetization represented in Fig. 11 was first observed (site 172, Table 1). The Cleaver Superchron should be recognizable globally, and it could become a global Proterozoic time marker in much the same way as the upper and lower limits of the 30 Ma long mid-Cretaceous Normal Superchron, or the 60 Ma Kiaman Reversed Superchron at the end of the Paleozoic are Phanerozoic global stratigraphic markers.

Syn-Hudsonian strata carry both polarities. Except for the Seton (all "normal") and Pearson (all "reversed") formations, both polarities always occur usually in unequal proportion (Table 3). Overall, however, "normal" and "reversed" are approximately evenly balanced—of the 295 horizons sampled from 18 rock units (1390 samples) 151 are "normal" and 175 "reversed". Evidently, during the interval approximately 1960–1830 Ma, the geomagnetic field spent approximately equal time in its normal and reversed states. The number of reversals is not known. In the Stark Formation, Bingham and Evans (1976) observed seven reversals, and in the thousands of meters of strata in the Coronation, Dubawnt, Goulburn, and Great Slave Supergroups, it is likely that scores, perhaps hundreds, are present. Though a daunting task, it should be possible to establish detailed magnetostratigraphic correlations of these four great sedimentary successions.

Can the true magnetic polarity be determined? This is an important question from geomagnetic and tectonic standpoints. According to Fig. 14, the present day northern part of Western Laurentia was in tropical latitudes during the interval ~1960–1830 Ma. Hildebrand (1988) and Hoffman and Grotzinger (1993) have reviewed estimates of paleo-trade wind directions from the elongation of stromatolite structures and other features. They suggested that paleowind blew predominantly from the south-southwest relative to the present frame. In the paleoframe of Fig. 14, this translates to wind from the east-northeast, broadly consistent with trade wind directions in a northern hemisphere location. It is strongly inconsistent with a southern hemisphere location.

In both basins there are local clockwise rotations about vertical axes of a few tens of degrees (Fig. 2, WR is exceptional) as determined paleomagnetically from stromatolite-bearing strata (Fig. 2). Inverting these rotations would change the inferred wind direction from south-southwest to south relative to the present frame. Relative to the paleoframe of Fig. 14, wind direction was from northeast, in excellent agreement with expected trade wind directions at a northern hemisphere location. As before, this is strongly inconsistent with a southern hemisphere location. Hence. the polarities designated "normal" and "reversed" seem to be truly normal and reversed in the Phanerozoic sense, the paleopoles of Fig. 12 are north geographic poles and Western Laurentia was in the northern hemisphere. This result also indicates that the time-averaged geomagnetic field was that of a geocentric axial dipole in Early Proterozoic time.

There remains the question of the true polarity of the Cleaver Superchron. If it is truly normal, then the changes in latitude between it and syn-Hudsonian time would be several times less than if it were reversed. On these grounds normal polarity is the preferred option; that is, paleopoles of Fig. 11 are north geographic poles, and Laurentia was in the northern hemisphere. These arguments mean that the forward connection across the Pacific Ocean, of the paleopoles of Figs. 11 and 12 to the Phanerozoic APW polar path for North America, as originally given by Irving and Park (1970), is correct.

In the Phanerozoic, the geomagnetic field has two stable states (Opdyke and Channell, 1996). One, during which reversals are common and the total duration of oppositely magnetized polarity chrons are statistically equal, as in the Cenozoic; the field recorded in syn-Hudsonian deposits had this general character. The second, during which reversals are very few or absent, and the geomagnetic field is strongly or exclusively biased towards one polarity; the field recorded by post-Hudsonian magnetization had this character. Hence, the reversal patterns for these two Early Proterozoic intervals are very similar to those observed in the Phanerozoic

9. Postscript

Within these two intervals, the earlier one of over 100 million years duration, and the later one probably of several tens of millions of years duration, the average directions of the geomagnetic field and the paleolatitudes deduced from them appears not to have changed in any systematic way. How did the field behave between them? It has been customary to connect paleopoles between these intervals by a looped path of apparent polar wander (APW). Reid et al. (1982) drew a shallow loop. Irving and McGlynn (1981) constructed a deep loop into the South Atlantic to accommodate paleopoles from elsewhere in Laurentia on the supposition that its major constituents were then assembled. Notably, they incorporated paleopoles from what they erroneously regarded as secondary magnetizations observed in the 2.2 Ga (Noble and Lightfoot, 1992) Nipissing Diabase (Symons and Londry, 1975; Roy and Lapointe, 1976) which is a large intrusive suite in the Southern Province on the southern margin of the Superior Province (Fig. 1). More recently, Buchan et al. (1989) have showed definitively from contact studies that these are not secondary but are truly primary magnetizations, and thus are far older than the APW loops constructed by Irving and McGlynn. Hence, the behaviour of the geomagnetic field over western Laurentia, in the interval recorded from syn-Hudsonian deposits and that recorded by post-Hudsonian magnetization, is not known, and there is now no reason to connect their corresponding paleopoles by a large-looped APW path.

Acknowledgements

We are pleased to acknowledge collaborations in the field with Robert S. Hildebrand and the late John C. McGlynn. We are grateful for help from Richard Franklin and Randy Enkin with diagrams. We are grateful to Pat Bickford for permission to use his new age for the Jan Lake Granite, and for his helpful discussion of its significance. We wish also to thank Sunil Gandhi who collected diabase from the Cleaver Dyke for dating. Finally, David Evans, R.S. Hildebrand, Vadim Krachinski, and Rob Rainbird provided reviews and comments which we greatly appreciate.

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